

COMP2005

Derivative Filters

In 1D

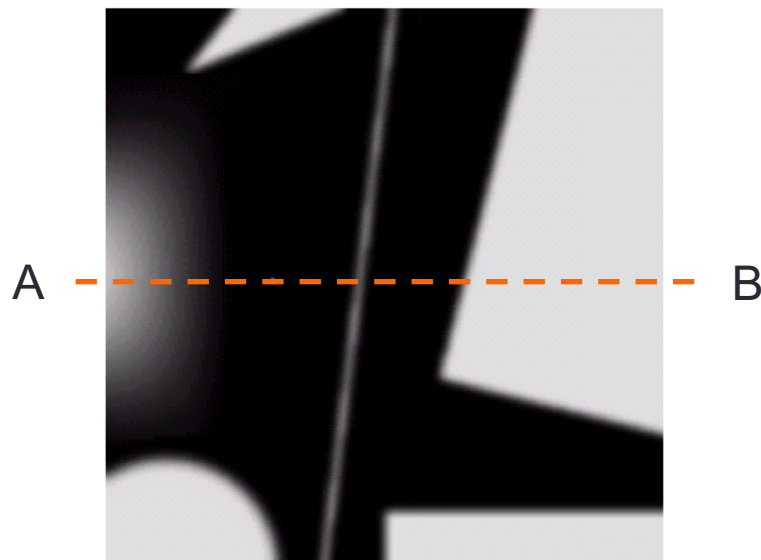
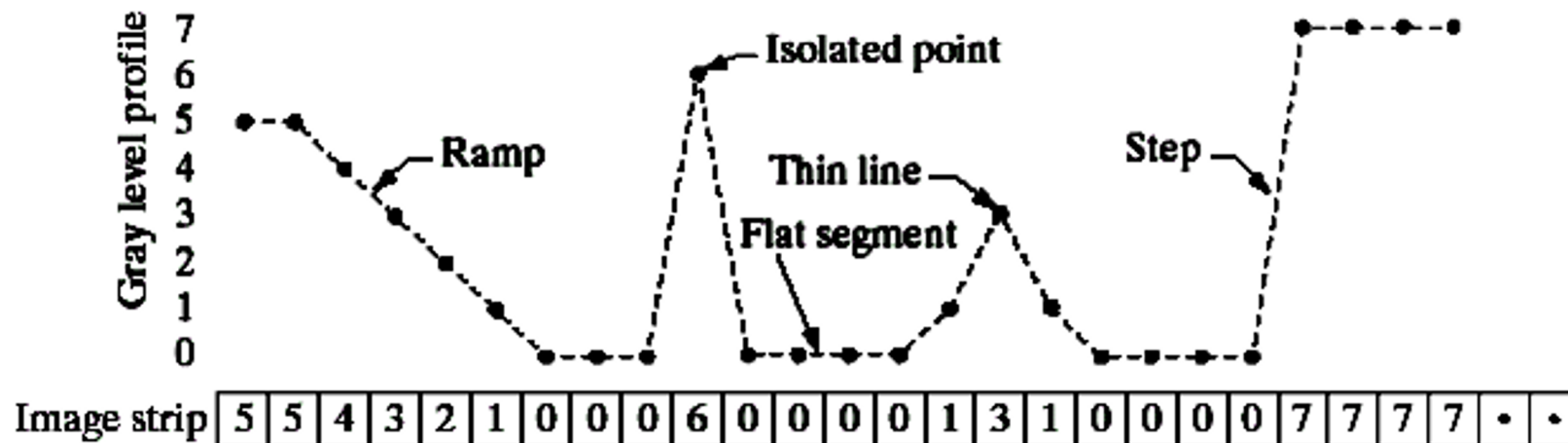


Image features are often characterised by changes in intensity

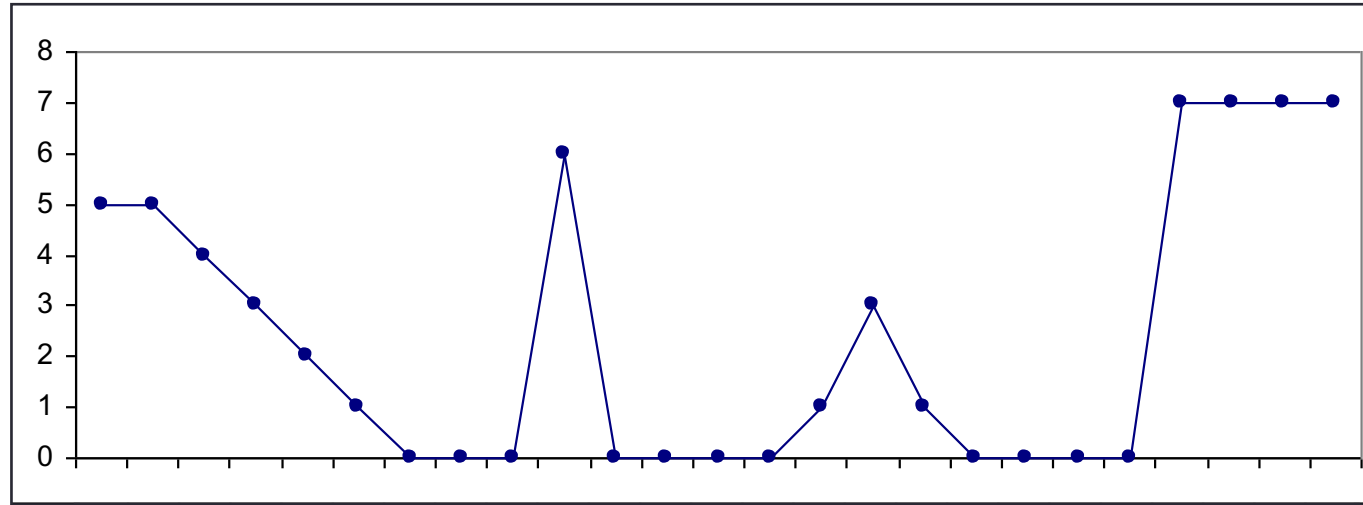


1st Derivative

- The 1st derivative of a function can be approximated by:

$$\frac{\partial f}{\partial x} = f(x+1) - f(x)$$

- The difference between neighbouring values and measures the rate of change of the function
- Can also be approximated by $(f(x+1) - f(x-1))/2$, etc.

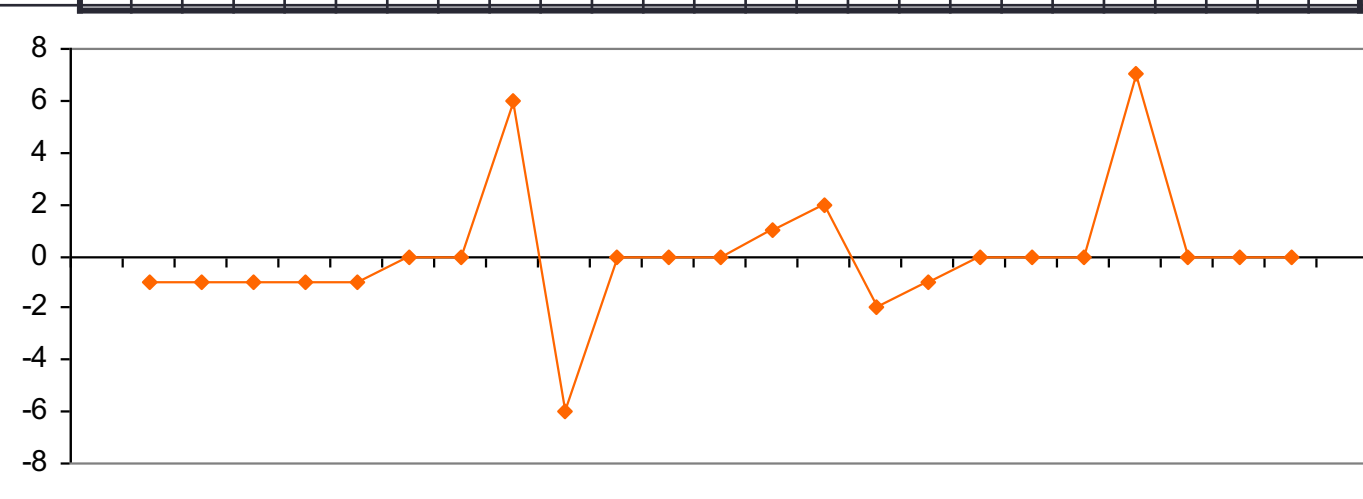


Raw data

5	5	4	3	2	1	0	0	0	6	0	0	0	0	1	3	1	0	0	0	0	7	7	7	7
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

	0	-1	-1	-1	-1	-1	0	0	6	-6	0	0	0	1	2	-2	-1	0	0	0	7	0	0	0
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1st
derivative



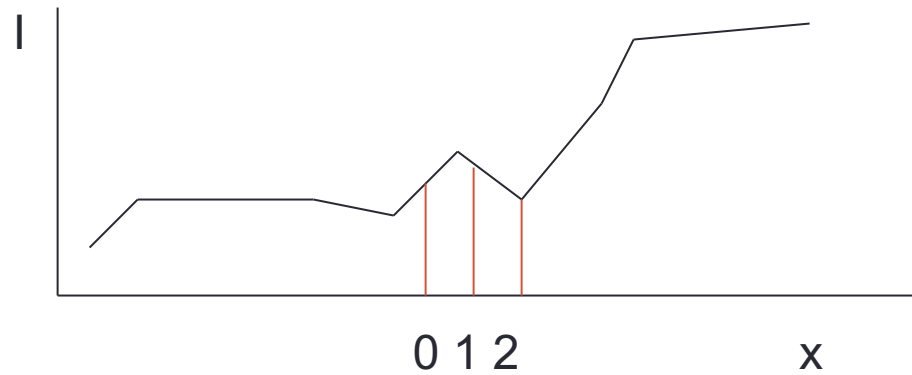
2nd Derivative

- The formula for the 2nd derivative of a function is:

$$\frac{\partial^2 f}{\partial^2 x} = f(x+1) + f(x-1) - 2f(x)$$

- Simply takes into account the values both before and after the current value
- Derived by estimating the 1st derivative at $x + 0.5$ and $x - 0.5$ and computing the derivative of the resulting data

2nd Derivative

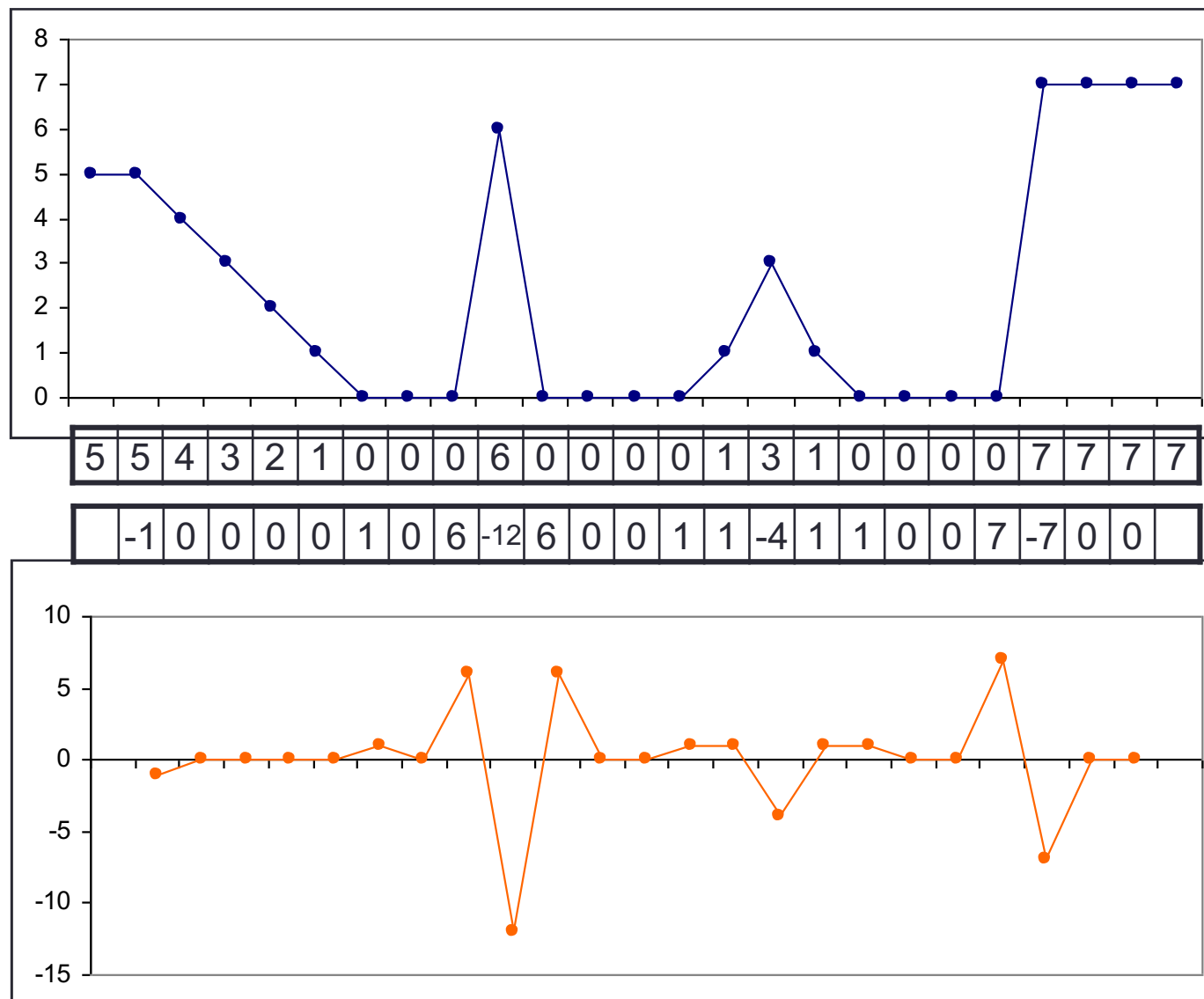


$$I''(1) = (I'(1.5) - I'(0.5))/1$$

$$I'(0.5) = (I(1) - I(0))/1 \quad \text{and} \quad I'(1.5) = (I(2) - I(1))/1$$

$$\therefore I''(1) = 1 \cdot I(0) - 2 \cdot I(1) + 1 \cdot I(2)$$

1	-2	1
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Raw data

2nd
derivative

Derivatives in 2D

- 2nd derivatives generalize to 2D quite easily, implementing a 1st derivative in 2D is a little more complex
- For a function $f(x, y)$ the gradient of f at coordinates (x, y) is given as the column vector:

$$\nabla f = \begin{bmatrix} G_x \\ G_y \end{bmatrix} = \begin{bmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \end{bmatrix}$$

- Computation of the 1st derivative can't be done by convolution alone

1st Derivative Filtering

- The magnitude of the 1st derivative vector is

$$\begin{aligned}\nabla f &= \text{mag}(\nabla f) \\ &= [G_x^2 + G_y^2]^{1/2} \\ &= \left[\left(\frac{\partial f}{\partial x} \right)^2 + \left(\frac{\partial f}{\partial y} \right)^2 \right]^{1/2}\end{aligned}$$

which can be simplified to

$$\nabla f \approx |G_x| + |G_y|$$

1st Derivative Filters

- Many 1st derivative filters have been proposed

- Roberts' Cross Operators

1	0
0	-1

0	1
-1	0

- Sobel Operators

-1	0	1
-2	0	2
-1	0	1

 G_x

-1	-2	-1
0	0	0
1	2	1

 G_y

These operators are most commonly associated with edge detection